CLIENT APPROVAL PACK

From Equipment QR to Trusted Service Record

Two-phase solution storyline for a faster, controlled and audit-ready field service operation

Prepared for	Document	Status	Date
PROMACH PTE. LTD.	Executive Solution Storyline	For Client Review & Approval	29 July 2026

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PURPOSE: This document converts the approved concept into a clear, phased delivery commitment. It is intended to support stakeholder alignment and formal client approval.

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1. Executive storyline

The requested solution is an equipment-driven field service platform. Its purpose is to remove repetitive technician data entry, ensure that every asset receives the correct maintenance activity, and produce a complete digital record that can be signed, distributed and traced.

The central operating principle is simple: the administrator configures each equipment record once, and the technician retrieves that configuration by scanning the equipment QR code. The technician records only what happened during the visit. The platform manages identification, checklist selection, ratings, normal limits, history, signatures, report generation and distribution.

PROPOSED DECISION: Approve the complete two-phase scope. Phase 1 establishes the mandatory QR-to-service workflow. Phase 2 adds advanced grouping, numbering, reporting, delivery controls and production hardening.

2. The business problem being solved

- Technicians currently risk repeating equipment details that already exist elsewhere.
- A generic checklist can cause the wrong maintenance activity or measurement fields to be used.
- Rated values may be re-entered incorrectly at each service visit.
- Paper or loosely structured reports make equipment-level history difficult to retrieve.
- Signatures, PDF copies and email evidence can become separated from the actual service record.
- Changes to checklists currently depend too heavily on system changes instead of administrator configuration.

The target state replaces these risks with a controlled data chain: one equipment master record, one unique QR identity, one automatically selected service form, and one auditable history.

3. The product promise

01	02	03	04	05
SCAN QR	AUTO-LOAD	SERVICE	SIGN	PDF & EMAIL

A technician should be able to move from one equipment item to the next with minimal interruption. The system identifies the equipment, applies the correct rules, records the work, and prepares the client-facing output.

Promise	What the user experiences	Business outcome
One QR, one equipment	A scan always opens the intended active equipment record.	Prevents asset selection errors.
Configuration, not coding	Admin changes checklists, parameters and limits through master data.	Faster operational change.
Minimal technician input	Only checklist results, actual readings and exceptions are entered.	Shorter service cycle and fewer errors.
Traceable completion	Every visit remains attached to the equipment history.	Stronger service evidence.
Controlled output	Signatures, PDFs and email logs remain linked to the same record.	Audit-ready client communication.

4. People and responsibilities

4.1 Administrator

The administrator owns configuration and control. Admin users maintain clients, locations, equipment specifications, QR codes, checklist templates, measurement parameters, limits, signature rules, recipient lists and report settings.

4.2 Field technician

The technician follows an intentionally narrow workflow. Identity is taken from the logged-in account; equipment and location are taken from the QR-linked master record; date and time are system generated. The technician completes the applicable work, enters actual readings, identifies exceptions and saves.

4.3 Client representative

The client representative reviews the applicable service output and, where the configured signature mode requires it, acknowledges the work using a secure link or the technician/admin device.

4.4 Management and operations

Management users review service history, completion status, abnormal readings, report distribution and audit evidence without modifying locked service results.

5. A day in the future workflow

5.1 Before the technician arrives

1. Admin registers the equipment under the correct client, site, building, level and room.
2. Admin records specifications, motor type, rated values, maintenance frequency and applicable limits.
3. Admin assigns the correct checklist and measurement parameters.
4. The system creates a unique QR code and printable equipment label.
5. The QR label is printed and fixed to the physical equipment.

5.2 At the equipment

1. The technician signs in and scans the QR code.
2. The system opens the exact equipment record and shows read-only identification and ratings.
3. The system loads the checklist for the applicable service frequency.
4. Only the selected measurement parameters and motor-phase fields are shown.
5. The technician completes checklist items and enters actual readings.
6. A reading stays Normal by default; an exception is highlighted through a clear tap or long-press action.
7. The technician adds remarks or compressed photos only when useful.
8. The technician saves the service record and scans the next equipment.

5.3 After the service activity

1. Completed equipment items are grouped according to the configured signature mode.
2. The relevant signature or no-signature completion rule is applied.
3. The platform generates the category or overall report number.
4. A professional PDF is produced and distributed to preset or manually entered recipients.
5. Sent date, time, recipients, signature details and report history are retained.

6. Two-phase delivery approach

PHASE 1	**Core QR-Enabled Service Operations** Deliver the mandatory equipment master, QR identity, dynamic service form, technician capture, history and baseline client output.

Phase 1 is the operational foundation. It proves the most important requirement: a technician can scan an equipment QR code and complete the correct service activity without re-keying equipment data.

- Secure administrator and technician access.
- Client, site, building, level and room/location hierarchy.
- Expanded equipment master with specifications, photos and lifecycle dates.
- Admin-only QR generator with unique, printable equipment labels.
- Configurable checklist templates by category and maintenance frequency.
- Configurable measurement parameters, units and normal limits.
- Single-phase and three-phase motor reading behaviour.
- Mobile scan-to-service workflow with autosave and validation.
- Normal-by-default readings with clear abnormal exception marking.
- Remarks and optional compressed photos.
- Equipment-level service history and audit trail.
- Baseline overall-report signature or email-only completion.
- Professional PDF and controlled email distribution.

PHASE 1 EXIT OUTCOME: A pilot technician can scan a real equipment label, complete the correct configured form, save the visit, obtain the applicable acknowledgement, generate the PDF and send it without manually entering equipment identity or rated values.

PHASE 2

Advanced Control, Distribution and Enterprise Readiness

Extend the proven field workflow with advanced signature grouping, category numbering, exception reporting, delivery evidence and operational controls.

- All configured signature modes: per equipment, selected equipment, category, overall report or no signature.
- Separate, concurrency-safe running numbers by equipment category.
- Advanced PDF grouping and exception presentation.
- Preset and manual recipient management with email audit and retry handling.
- Correction, revision and locked-report controls.
- Operational dashboards, search, filters and service-history views.
- Security hardening, backup/restore, monitoring and support runbooks.

- Pilot refinement, user training, controlled rollout and acceptance sign-off.

PHASE 2 EXIT OUTCOME: The platform supports the client's full signature, numbering, reporting and distribution variants and is ready for controlled production operation.

7. Experience principles

Principle	Design commitment
Scan first	The primary field entry point is the equipment QR code, not a manual equipment form.
Read-only master facts	Technicians can see equipment facts but cannot alter them during a visit.
Show only what applies	Checklist items, measurement parameters and phase fields are filtered by configuration.
Exceptions deserve attention	Normal is quiet; abnormal conditions are visually prominent and traceable.
Touch-friendly	The service form uses large controls, clear progress and minimal navigation on phones.
Recoverable work	Draft progress is preserved so a temporary interruption does not lose the visit.
Signed means locked	A signed or email-only completed record cannot be silently changed.
History follows equipment	Each future visit adds to one continuous equipment service history.

8. What success will look like

- No technician manually types client, location, tag, brand, model, serial number or rated values during a normal QR-led visit.
- The scanned equipment always opens the configured checklist and selected measurement fields.
- Out-of-range or manually marked abnormal readings are easy to identify and review.
- Every completed visit can be found from the equipment history.
- Every issued report can show who completed, signed and received it, with timestamps.
- Admin users can change checklist content and parameter assignments without a software deployment.
- Mobile field use remains clear and efficient on supported devices.

9. Controls that protect the operation

The solution will use role-based access, secure sessions, opaque QR identifiers, server timestamps, validated uploads, locked completed records and audit events. QR values will not expose raw database identifiers or confidential equipment details.

Existing service records will retain the exact checklist version, parameter set, readings and signature rule that applied at completion, even if an administrator later changes the master configuration.

IMPORTANT BOUNDARY: AI recommendations, GPS attendance, offline synchronization, inventory, spare parts, scheduling, billing, invoicing, IoT integration and WhatsApp are not included in this approval unless added through formal change control.

10. Decisions requiring client confirmation

Decision	Recommended position	Why approval is required
PDF measurement rule	Show all selected readings and highlight abnormal values; add an exception summary.	The request can also be read as omitting all normal readings.
Frequency behaviour	Allow admin to define replacement or cumulative templates.	Quarterly work may include or replace monthly items.
Abnormal detection	Warn automatically outside limits and require technician confirmation.	A manual-only red mark can miss genuine exceptions.
QR fallback	Allow audited admin-authorized search only when a label is damaged.	The normal technician flow remains scan-only.
Number reset	Client to select daily, yearly or continuous sequence per category.	The example does not define the reset rule.
Email size handling	Compress toward 10 MB; use a secure link if the file remains larger.	A strict size guarantee is not possible for every photo set.
Retention and hosting	Confirm policy, region and backup expectations before production.	These affect architecture and operating cost.

11. Approval requested

Client approval of the companion Functional Scope and Approval Document confirms the proposed two-phase commitment, stated business rules, boundaries and acceptance approach. Commercials, detailed delivery dates and service-level commitments will be issued or confirmed separately after scope approval.

NEXT STEP: Review both documents, record answers to the decision items, and sign the approval section in the Functional Scope and Approval Document.